A1 Week 4 Plan

Role 1 (Optimal Contract and General Carbon Theory)

- **Prove the carbon classification theorem.** Construct a single distributional functional of the residual-demand law, and prove the sign of the net-carbon effect equals the sign of that functional – both directions. This turns the Week-2 sufficient conditions and numerical resolution into one closed-form if-and-only-if: exactly when, and why, a flexible load reduces carbon, for every distribution. It is the headline result, and it must be airtight to the most adversarial reviewer.
- **Pin the minimal decisive statistic.** Week 2 proved the carbon sign is not decidable from the distribution-free cutoffs, so the functional must live in the right conditional-moment space. Identify that minimal sufficient statistic and prove the converse – no coarser statistic of the distribution decides the sign – so the characterization is not just sufficient but sharp.
- **Close the frontier leverage condition to a theorem.** The Week-3 extension reduced global carbon-optimality of the surplus contract to a single leverage condition, verified but not proved. Prove it – or characterize exactly the class of distributions and costs on which it holds and the boundary where it fails – completing the optimal-contract-family result.
- **Give the distributionally-robust characterization.** Combine the functional with an ambiguity set of demand laws: characterize the worst-case distribution and the robust-optimal contract in terms of the functional, with a closed form on the power-law family and a price of robustness.
- **Draw the carbon phase diagram from the functional.** Partition the primitive space (costs and distribution shape) into carbon-up, carbon-neutral, and carbon-down regions whose analytic boundaries are the level sets of the functional and the viability threshold; establish the topology (each region connected, boundaries monotone).
- **Unify and self-contain.** Assemble the above into one classification theorem and make the deliverable standalone, restating and proving every imported lemma in-file. The numerical companion mirrors every identity and confirms sign of carbon equals sign of the functional in every cell.

Role 2 (Solver and Numerical Methods)

- **Build the classification-functional evaluator.** Compute the functional across a large theory-tied distribution library (Beta scale-mixtures, bimodal, heavy-tailed) and confirm the sign of the net-carbon effect matches the sign of the functional in every cell, to machine precision – one principled sweep that corroborates the if-and-only-if or surfaces the counterexample.
- **Map the phase diagram numerically.** Trace the level sets of the functional and the viability threshold across the primitive space, rendering the carbon-up / neutral / down partition and certifying the boundaries (not finite-difference noise).
- **Certify the leverage and the robust saddle.** Verify the leverage condition and its boundary numerically, and certify the distributionally-robust saddle (worst-case distribution matches the inner problem; minimax equals maximin).
- **Delimit the regularity region.** Establish the boundary of where the characterization's hypotheses hold (analytic Hessian / KKT certificates), delivering a numerical map of where

the Week-4 theory applies.

Role 3 (Comparative Statics and Robustness)

- **Sign the functional's comparative statics.** Via implicit-function and envelope identities, sign the derivative of the classification functional and the movement of the carbon boundary for every primitive, and prove the direction in which each cost shifts the carbon-down region.
- **Quantify the value of robustness through the functional.** Prove monotonicity of the robust-minus-nominal carbon gap in the ambiguity radius, bound the price of robustness, and deliver a cross-distribution certificate along a likelihood-ratio / mean-preserving spread path so the verdict holds across the path.
- **Map fragility of the boundary.** Identify where the characterization is knife-edge – the level set of the functional at zero – and construct explicit witnesses on each side, the sharpness companions to Role 1's theorem.
- **Sign the contract-family comparative statics.** Characterize how the optimal budget and the leverage move with costs and with the shape of the distribution, in particular near the boundary where surplus-only ceases to be globally optimal.

Role 4 (Theory Hardening and Regimes)

- **Extend the functional across regimes.** Carry the classification functional into the small-FCL regime and across the regime boundary; prove the functional and the carbon boundary stitch continuously, or locate and explain the discontinuity.
- **Harden the converse.** Supply the measure-theoretic scaffolding the necessity half needs – existence and measurability of the extremal laws, constraint qualification for the minimal-statistic argument, and the class of admissible distributions on which the converse is tight.
- **Regime-map the classification.** Determine where the carbon regions and their boundaries cross regime lines; deliver the stitched phase diagram over the full cost plane spanning both FCL regimes.
- **Reconcile the team.** Integrate the four roles' Week-4 outputs into one coherent statement, cross-check every shared object (the functional, the boundary, the robust optimum) for consistency across roles, and flag any divergence before it reaches the integrated paper.